CHALLENGE #7

1. Download Dataset

I used python programming language with Google Colab platform, so the dataset from kaggle can be directly accessed because it is open source. This dataset consists of two files, namely beer data and breweries data.

2. Join two data

These two datasets have one column/feature that is related from the scraping results, namely 'brewery_id'. Here I use the left join technique from pandas python.

```
df_merged = pd.merge(df_beer, df_breweries, on='brewery_id', how='left')
df_merged.head()
```

```
df_merged = pd.merge(df_beer, df_breweries, on='brewery_id', how='left')
df_merged.head()
```

	Unnamed: 0	abv	ibu	id	name_x	style	brewery_id	ounces	name_y	city	state
0	0	0.050	NaN	1436	Pub Beer	American Pale Lager	408	12.0	10 Barrel Brewing Company	Bend	OR
1	1	0.066	NaN	2265	Devil's Cup	American Pale Ale (APA)	177	12.0	18th Street Brewery	Gary	IN
2	2	0.071	NaN	2264	Rise of the Phoenix	American IPA	177	12.0	18th Street Brewery	Gary	IN
3	3	0.090	NaN	2263	Sinister	American Double / Imperial IPA	177	12.0	18th Street Brewery	Gary	IN
4	4	0.075	NaN	2262	Sex and Candy	American IPA	177	12.0	18th Street Brewery	Gary	IN

We can see that the two datasets have become one dataset. Some column names changed from the previous data, namely 'name' in the beer dataset to 'name_x' (beer brand name) and 'name' in the breweries dataset to 'name_y' (breweries name).

3. Discover cities and Breweries to visit

- ***The two most common styles of beers on the list***

By counting the number of unique values of each beer style.

```
style_counts = df_merged['style'].value_counts()
two_most_common_styles = style_counts.head(2)
print(two_most_common_styles)
```

It was found that the two most common beer styles were **American IPA** with 424 and **American Pale Ale (APA)** with 245.

- ***The city which produces most of brands of beers (from two common styles of beers)***

By filtering the dataset using the keywords American IPA and Americam Aple Ale (IPA) then calculating the unique value of grouping city names and beer brands using the groupby method in python. Here is the code;

```
american_ipa_df = df_merged[df_merged['style'] == 'American IPA']
american_apa_df = df_merged[df_merged['style'] == 'American Pale Ale (APA)']
city_brand_counts_ipa = american_ipa_df.groupby('city')['name_x'].nunique()
city_brand_counts_apa = american_apa_df.groupby('city')['name_x'].nunique()
print(city_brand_counts_ipa.idxmax())
print(city_brand_counts_apa.idxmax())
```

It was found that the cities that produce the most beer brands of the two styles are **Tampa** and **Chicago**.

- ***The most common styles of beers produced for particular state (number of brands produced, average ABV and IBU)***

Firstly, I'm looking at the two states that produce the most types of beer brands: California (CA) and Colorado (CO). I then analysed to find the most common beer styles produced from these two countries and looked for information related to the number of brands produced and also the average ABV and IBU of these beer styles.

Colorado (CO)	California (CA)
style American IPA 40 American Pale Ale (APA) 40 Name: count, dtype: int64 Style: American IPA Number of Unique Beer Names: 36 Average IBU: 70.34782608695652 Average ABV: 0.06613157894736842 -------------------- Style: American Pale Ale (APA) Number of Unique Beer Names: 37 Average IBU: 56.4 Average ABV: 0.05876923076923076 --------------------	style American IPA 45 American Pale Ale (APA) 17 Name: count, dtype: int64 Style: American IPA Number of Unique Beer Names: 42 Average IBU: 66.6842105263158 Average ABV: 0.06711111111111111 -------------------- Style: American Pale Ale (APA) Number of Unique Beer Names: 16 Average IBU: 40.92307692307692 Average ABV: 0.05152941176470586 --------------------

- ***The city or state which produced the most distinct styles of beers.***

The most distinct type of beer style means a rare or one-of-a-kind style. Here, we first look for unique beers. Then, we make all types of beer styles into a list dictionary based on the country. After that, using a for loop, we search for beer styles from a country that are not owned by other countries.

	abv	ibu	id	name_x	style	brewery_id	ounces	name_y	city	state
2050	0.063	23.0	1366	Stupid Sexy Flanders	Flanders Oud Bruin	25	16.0	Sun King Brewing Company	Indianapolis	IN

The search results show that the rarest beer style is Flanders Oud Bruin from the state of Indiana, city of Indianapolis.

- ***Top 5 cities and breweries wish list.***

From all of this information, the 5 best cities and 5 best breweries that I will visit are:
City:
- Tampa (City with the most American IPA beer styles)
- Chicago (City with the most American Pale Ale (APA) beer styles)
- Indianapolis (City with the most distinct beer styles with only one, Flanders Oud Bruin.
- Portland (the city with the most breweries)
- Grand Rapids (the city with the most beer brands)

Breweries:
- Brewery Vivant (Breweries with the most brand beer production of all breweries)
- Oskar Blues Brewery (Breweries with the most brand beer production in Colorado)
- 21st Amendment Brewery (Breweries with the most brand beer production in California)
- Portside Brewery (Breweries with the highest ABV)
- Sun King Brewing Company (Brewer of the rarest Flanders Oud Bruin beer style)